

DESIGNING 3-PHASE 4-POLE SQUIRREL CAGE INDUCTION MOTOR

Course No: EEE 206

Course Title: Energy Conversion Lab-I



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Bangladesh University of Engineering and Technology

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A Report On

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Forwarding Letter

December 9, 2024

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Subject: Report submission on a project entitled “*Designing a 3-Phase 4-Pole Squirrel Cage Induction Motor*” of course no. EEE 206 (Energy Conversion Laboratory)

Dear Teachers,

We are pleased to collectively present the group report titled “Designing a three phase four pole Squirrel Cage Induction Motor” for your thoughtful review. This collaborative effort, under your esteemed guidance, endeavors to explore the construction of an induction motor and rewind it following all criteria. Our analysis not only underscores our prime objective, which is to count number of turns from motor ratings, but also covers the determination of circuit parameters and highlights the performance of the motor on the basis of torque-speed characteristics.

We extend our sincere gratitude for your invaluable guidance throughout this process, as well as our appreciation to those who contributed to the completion of this project. The insights gathered from the questionnaire and distributed among our group mates have been instrumental in enriching our analysis. We recognize the collaborative efforts of our team in compiling this report and approach this task with dedication.

And finally, we humbly apologize for any unintentional mistake we have committed in this process. We hope that you would be kind enough to pardon us for those mistakes and oblige thereby.

Sincerely,

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Objectives

- Designing a squirrel cage induction motor
- To obtain the motor equivalent circuit parameters by performing necessary tests.
- To observe the torque vs. speed characteristics of the motor.
- Performance Analysis

2. Introduction

Induction motor is a widely used motor from household to industrial applications. It is essentially made up of a rotor that can rotate continuously around its axis and a stationary stator. The motor's stator receives a three-phase voltage, which creates a revolving magnetic field. Consequently, a current flows and a voltage is induced on the rotor by this magnetic field. Additionally, this current generates a rotor magnetic field, which interacts with the stator magnetic field to produce a net torque that generates rotation.

An induction machine is essentially a device that has a single continuous set of amortisseur windings. Because the rotor voltage, which generates the rotor current and magnetic field, is induced in the rotor windings rather than being physically connected by wires, these devices are known as induction machines. An induction motor's unique selling point is that it doesn't require a dc field current to operate.

Even though an induction machine can be used as a generator or a motor, it is only utilized as a generator in specific situations due to its numerous drawbacks. Because of this, induction machines are typically called induction motors.

Induction motors come in two varieties. The squirrel cage induction motor and the wound rotor induction motor are the two types. In this project, we will be working with a three-phase squirrel cage induction motor.

3. Motor Specifications

The purchased induction motor's specifications on the nameplate are given in the following table:

Power	0.18kW	Insulation Class	F
Rated Speed	1400rpm	Power Factor	0.70
Rated Voltage	415V	Efficiency	57%
Rated Current	0.73A	Ingress Protection Code	55
Frequency	50Hz		



Fig 1: Nameplate of motor

4. Construction

The construction of motor can be discussed of the following parts-

- Stator
- Squirrel cage rotor
- Casing
- Cooling Fan

Stator:

There are 24 slots in the stator. The motor was designed to have 24 slots, fractional pitch, and single layer distributed winding. Therefore, in order to design the stator, 12 coils were required. Copper wire was used to create the coils. The stator receives a three-phase power source and generates a four-pole rotating magnetic field. Insulation was positioned between the coils and a ferromagnetic core. Class F insulation, which can withstand a maximum temperature increase of 105°C from the ambient temperature of 40°C, was selected. The insulation consists of materials or combinations of materials such as mica, glass fiber, asbestos, etc., with suitable bonding, impregnating, or coating substances, as well as other materials or combinations of materials, not necessarily inorganic. It protects the ferromagnetic core from overheating.



Fig 2 : Construction of stator

Rotor:

The rotor does not have any windings, instead it has some conducting bars shorted by a large shorting ring. The conducting bars are placed inside laminated iron core. The bars are skewed to reduce slot harmonics. Some fan blades come out of the rotor to cool down the rotor. This is the main rotating part of the motor and is connected to the shaft. The rotor design class is A.



Fig 3: Squirrel cage rotor

Casing:

It is the outside cover of the motor made of metal. It supports and protects the internal components (stator and rotor).

Cooling Fan:

A cooling fan is connected with the rotor. Hence, the cooling fan rotates with the rotor and dissipates the heat generated. It protects the motor from overheating.



Fig 4 : Cooling fan

5. Winding Type and Calculations

- **Winding Type:**

Chosen Winding Type	4-pole Fractional Pitch Single-Layer Distributed Winding
Connection Type	Y

- **Pitch Factor Calculation:**

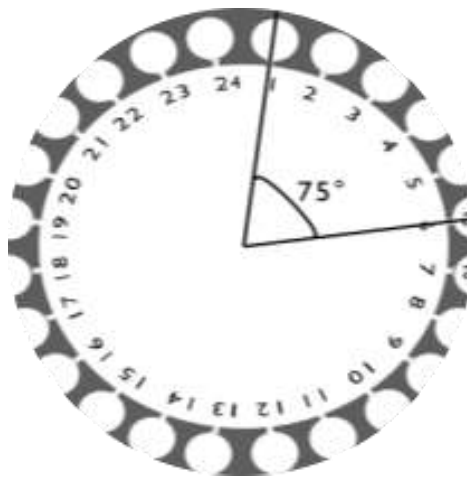


Fig 5: Coil Pitch

Coil Pitch(Mechanical)	75°
Coil Pitch(Electrical), ρ	$75^\circ \times 2 = 150^\circ$
Pitch Factor, K_p	$\sin\left(\frac{\rho}{2}\right) = 0.9659$

- **Distribution Factor Calculation:**

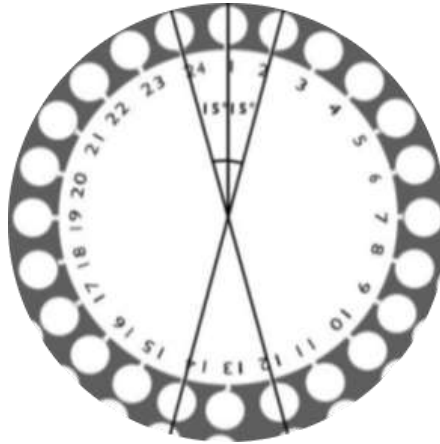


Fig 6: Slot pitch and phase belt

Slots per pole	$\frac{s}{p} = \frac{24}{4} = 6$
Slots per phase belt	$N_{sl} = \frac{s}{pm} = \frac{24}{4 \times 3} = 2$
Electrical Angle per slot	$\gamma = \frac{180 \times p}{s} = \frac{180 \times 4}{24} = 30^\circ$
Distribution factor	$K_d = \frac{\sin(\frac{N_{sl} \times \gamma}{2})}{N_{sl} \times \sin(\frac{\gamma}{2})} = 0.9659$
Angle covered by coil	$\theta_m = 75^\circ$
Pitch factor	$K_p = \sin(\frac{p \times \theta_m}{4}) = 0.966$
Winding factor	$K_w = K_p \times K_d = 0.933$

- **Flux Per Pole Calculation:**

Slot Length, L	6.6cm
Diameter, D	5.7cm
Number of Poles, P	4
Average Flux Density, B_{av}	0.5T
Flux Per Pole, ϕ	$\frac{B_{av} \times \pi \times D \times L}{p}$ $= \frac{0.5 \times \pi \times 0.057 \times 0.066}{4}$ $= 1.48 \times 10^{-3} Wb$

- **Turns and Wire Size Calculation:**

Turns Per Phase, N_p	$\frac{E}{\sqrt{2}\pi f \phi K_w} \approx 783$
Turns Per Coil, N_c	$\frac{N_p}{i} = \frac{N_p}{4} = 195.63 \approx 196$
Wire Cross Section, A	$A_s = \frac{I_L}{\delta_s} = 0.104 \text{ mm}^2$ (29 SWG)

Here,

Phase Voltage, $E = \frac{400}{\sqrt{3}} = 230.94$; (Y Connected)

Rated Current, $I_{\text{rated}} = 0.73$ A

Current Density, $\delta_s = 7 \text{ A/mm}^2$ (assumed for low voltage small machines ^[2])

Number of coils per phase, $i = 4$

Standard Wire Gauge	Diameter		Turns of wire		Cross-sectional area		Res. per length (for copper wire)		Mass per length		Current Capacity / A	
	in	mm	in ⁻¹	mm ⁻¹	kcmil	mm ²	Ω/kft	Ω/km	lb/ft	kg/m	750 kcmil/A	500kcmil/A
25	0.0200	0.508	50.0	1.97	0.400	0.203	85.1	279	0.00121	0.00181	0.533	0.800
26	0.0180	0.457	55.6	2.19	0.324	0.164	105	345	984 μ	0.00146	0.432	0.648
27	0.0164	0.417	61.0	2.40	0.269	0.136	127	415	817 μ	0.00122	0.359	0.538
28	0.0148	0.376	67.6	2.66	0.219	0.111	155	510	665 μ	990 μ	0.292	0.438
29	0.0136	0.345	73.5	2.89	0.185	0.0937	184	604	562 μ	836 μ	0.247	0.370

Fig 7: Part of Standard Wire Gauge Table

- **Code Letter Determinatuion:**

Line Voltage, $V_L = 415\text{V}$

Starting current (According to website^[3]), $I_{\text{Start}} = 0.73 \times 500\% = 3.65\text{A}$

$S(\text{start}) = \sqrt{3} \times V_L \times I_{\text{Start}} = 2.624\text{kVA}$

Rated HP= 0.25HP

Code Letter Factor = $\frac{S(\text{start})(\text{kVA})}{\text{Rated HP}} = 10.49$

So the code letter is “**M**” (10 -11)

6. Winding Diagram

Ours is a 4-pole motor where we have 24 slots. We shall use single layer winding and so, we shall need 12 coils. Also, we shall use fractional pitch winding. Angle covered by each coil would be 75° . Therefore, if a coil starts at slot number 1 it will end at slot number 6.

The picture inserted above is a rough picture but will be very much helpful in our analysis. We have numbered the coils in CCW direction and now we will number the slots.

Coil	Slot		Coil	Slot	
	Start	End		Start	End
1	1	6	7	13	18
2	3	8	8	15	20
3	5	10	9	17	22
4	7	12	10	19	24
5	9	14	11	21	2
6	11	16	12	23	4

Let's take Phase A into consideration.

Phase A starts from coil 1. We shall start from slot 1, which also marks the beginning of coil 1. Then we move on to coil 4 and we short the lower heads that is slot 6 and slot 12. After that we go to coil 7 and we short the upper heads which are slot 7 and 13. Finally, we move on to coil 10 and we short the lower heads 18 and 24. So, we have upper head of coil 1 which is slot 1 and upper head of coil 10 which is slot 19 open and we shall provide one phase of ac supply there.

Therefore, the order of slots for A phase is:

1-----→6 (Coil 1)
↓
 7 ←-----12 (Coil 4)
 ↓
 13-----→18 (Coil 7)

.....↓
19←-----24 (Coil 10)

Similarly, the order of slots for B phase is:

5-----→10 (Coil 3)
.....↓
11 ←-----16 (Coil 6)
↓
17-----→22 (Coil 9)
.....↓
23←-----4 (Coil 12)

Similarly, the order of slots for C phase is:

9-----→14 (Coil 5)
.....↓
15 ←-----20 (Coil 8)
↓
21-----→2 (Coil 11)
.....↓
3←-----8 (Coil 2)

And finally it will look like this:

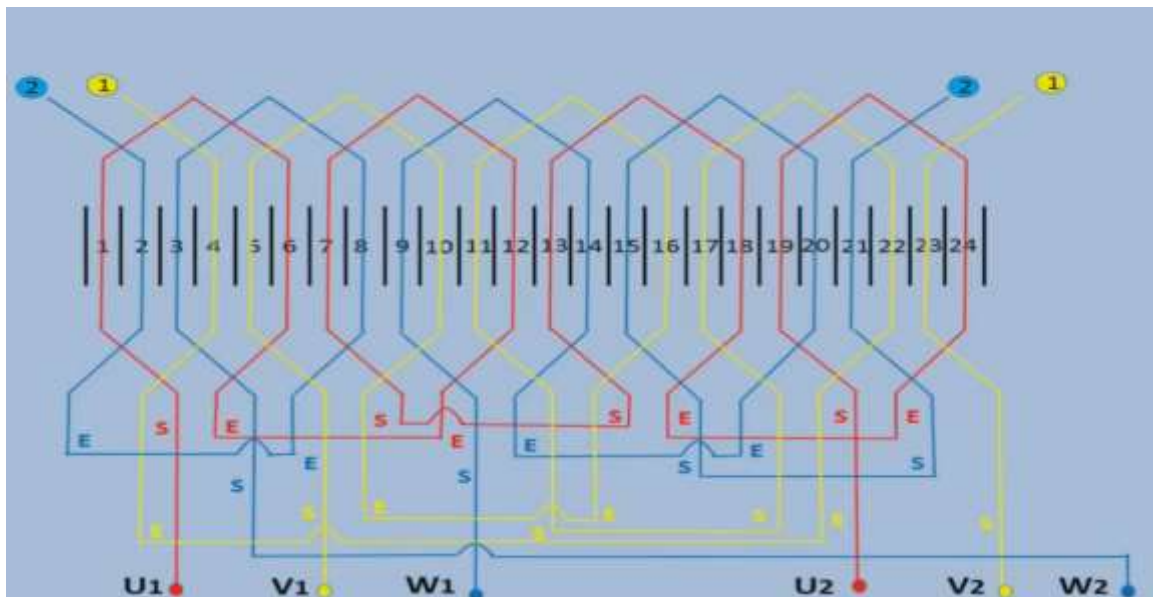


Fig 8: Winding Diagram of Three Phase Four Pole Single Layer winding

7. Wire Selection

Options:

- Copper
- Aluminum
- Silver

Copper	Aluminum	Silver
<u>Advantages:</u> ▪ Very high conductivity (5.96×10^7 S/m), so wire cross section area will be smaller for a certain current which will lower material cost. Moreover, the volume of stator will be small. ▪ Low cost ▪ Availability	<u>Advantages:</u> ▪ Lightweight (Density of 2.7g/cm^3). This will lower the weight of the motor. ▪ Low cost. ▪ Availability ▪ Environment friendly	<u>Advantages:</u> ▪ Very high conductivity (6.3×10^7 S/m). Higher than both copper and aluminum.
<u>Disadvantages:</u> ▪ High density (8.96g/cm^3). So, the motor weight will increase. ▪ Toxic, affects plant growth in soil, damages kidney and liver when consumed.	<u>Disadvantages:</u> ▪ Comparatively lower conductivity (3.6×10^7 S/m), so wire cross section area will be larger than that of coppers. ▪ Higher cost. ▪ Increased volume of the stator	<u>Disadvantages:</u> ▪ Substantially high cost. ▪ Not easily available

Conclusion:

The best performance at the lowest cost is offered by copper wire. However, it is impossible to overlook its impact on the environment. Copper is used in the windings because it is the only suitable material.

8. Insulation

When current flows through a wire, there will be an energy loss due to the I^2R effect. This energy is lost in the form of heat energy. Therefore, the machine will be heating up when it is being operated. To make sure that there is no serious damage done to the motor during operation at rated condition, engineers use various types of insulation coating to insulate the ferromagnetic material core from overheating. Depending on the temperature that they can withstand, there are several classes.[2]

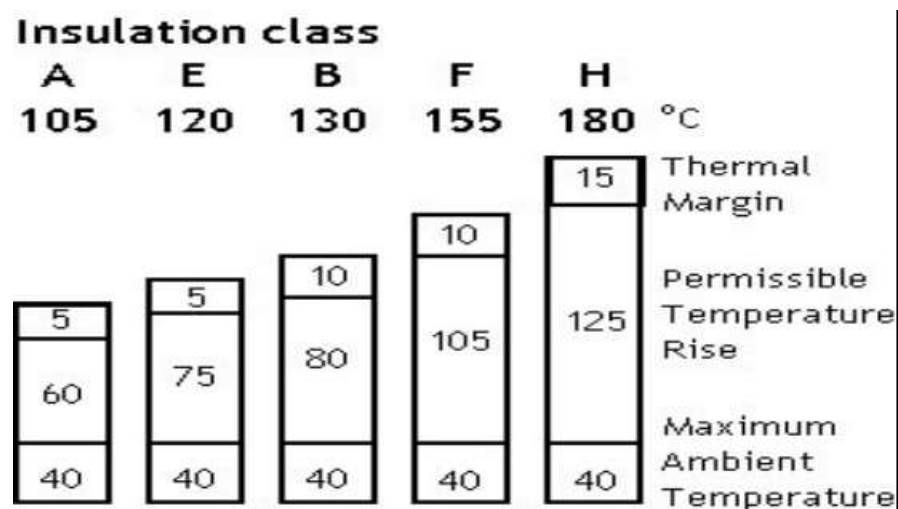


Fig 9: Various insulation classes[7]

The maximum allowable temperature for each insulation class is determined by their maximum permissible temperature rise with respect to the maximum ambient temperature of 40° C.

According to a rule of thumb, for each 10° C temperature rise from the threshold temperature, the machine's lifetime cuts by half. The graph below confirms this empirical relation between the lifetime of a machine and the temperature rise.

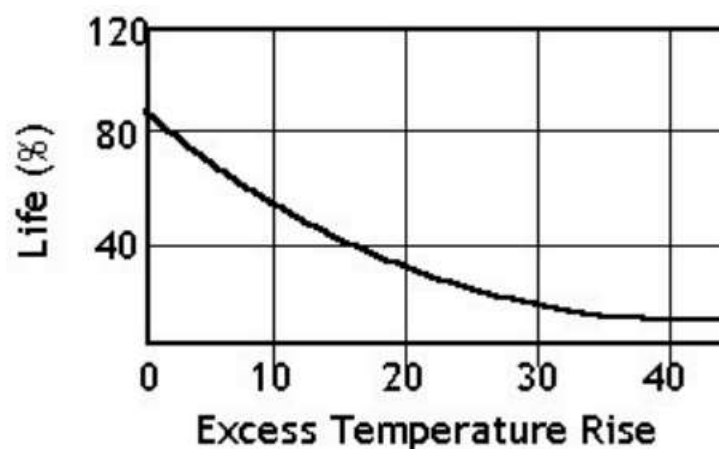


Fig 10: Lifetime expectancy curve vs temperature rise above threshold

There are three factors for choosing insulation class-

- Safety
- Cost
- Lifetime

Longer lifespan and improved safety are provided by higher insulation classes. But it comes at a higher price as well. Class "F" is the insulation class that we have selected for our induction motor because we have given the machine's longevity and safety top priority. The insulation class "F" is capable of withstanding a temperature increase of 105°C over room temperature. Therefore, we can say that it is fairly safe for our machine.



Fig 11 : Fish paper Insulation



Fig 12 : Polyester Insulation



Fig 13 : Insulation Placed in slots

Fish Paper Insulation:

Fish paper – or vulcanized fiber – is a durable and flexible type of fibrous electrical insulation paper. Fish paper is also unique because it is more resistant to heat and cold

comparable plastic materials. Electrical-grade Fish paper is a vulcanized fiber material that is frequently used for electrical insulation and is commonly die cut to create unique parts for specific applications.

Polymer Insulation;

Since polyester insulation is composed of 45% recyclable plastic, it is environmentally friendly. It has a durability rating of over fifty years. It can withstand heat and is waterproof. It can be applied around the stator slots as an insulating substance.

Varnishing;

Varnishing the windings of an electric motor or generator functions to insulate the windings from contaminants, to make the windings rigid and tight, and to dissipate heat. Increasing the efficiency of the machine.

9. Constructing the Designed Motor

1. Opening The Motor : At first the screws were opened and then the fan cover came out as it was pulled. It exposed the cooling fan which was also separated similarly. Then the end bell was hit with a hammer and the rotor came out through the other side. It also exposed the stator coil.



Fig 14: Rotor with shaft and exposed stator coil

2. Removing Stator Coil : The stator coils were cut from one side and it came out as it was pulled from the other side .



Fig 15 : Pulled out stator coil

3. Putting Insulator : After removing the coil motor slots were cleaned. Then class F insulator was cut according to the size of the motor slot and placed inside.



Fig 16: Putting insulator

4. Making Stator Coil : 24 coils were made by using copper . Each coil contained 200 turns. The coils were made using a tool which allowed the coils to have the same size as the stator slots.



Fig 17 : Making new coil

5. Placing Coils : There were 24 slots for 4 coils. We placed a coil in two slots keeping three slots empty in between and also kept one slot empty between two coils to ensure no short.



Fig 18: Coil placing

6. Connecting Coils : At first insulators were removed from the extension of the coils kept for the connection. Then coils were connected using these extensions in 3 phase 4 pole sequence.



Fig 19: Coil Connection

7. Re-assembling The Motor : The rotor was then placed inside the motor . The cases were put accordingly, fan was connected and the whole setup was sealed with screws.

10. Equivalent Circuit

1) DC test:

$$R_{ab} = 64.2\Omega$$

$$R_{bc} = 64.3\Omega$$

So the value of stator resistance,

$$R_1 = \frac{R_{ab} + R_{bc}}{4} = 32.125\Omega$$

2) No Load test:

Obtained data,

$$V_T = 400V$$

$$I = 1.3A$$

$$P = 197W$$

$$X_1 + X_M = \frac{V_T}{\sqrt{3}} \div I$$

$$\rightarrow X_1 + X_M = \frac{400}{\sqrt{3}} \div 1.3 = 177.64\Omega$$

3) Blocked Rotor test:

Obtained data,

$$V_T = 80.3V$$

$$I = 0.73A$$

$$P = 557W$$

$$Z_{LR} = \frac{V_T}{\sqrt{3}} \div I$$

$$Z_{LR} = \frac{80.3}{\sqrt{3}} \div 0.73 = 63.51\Omega$$

$$\theta = \cos^{-1} \frac{P}{\sqrt{3} \times I \times V_T}$$

$$\theta = \cos^{-1} \frac{557}{\sqrt{3} \times 80.3 \times 0.73} = 0.73$$

$$R_1 + R_2 = Z_{LR} \times \cos \theta = 46.36\Omega$$

$$X_1 + X_2 = Z_{LR} \times \sin \theta = 43.40\Omega$$

As our motor is of Class B,

$$X_1 = 0.6 \times 43.40 = 17.36\Omega$$

$$X_2 = 0.4 \times 43.40 = 26.04\Omega$$

Then,

$$X_1 + X_M = 177.64\Omega$$

$$17.36 + X_M = 177.64\Omega$$

$$X_M = 160.28\Omega$$

Finally,

$$R_1 + R_2 = 46.36\Omega$$

$$32.125 + R_2 = 46.36\Omega$$

$$R_2 = 14.235\Omega$$

Having determined the circuit parameters, we can now draw the equivalent circuit in PSpice schematics.

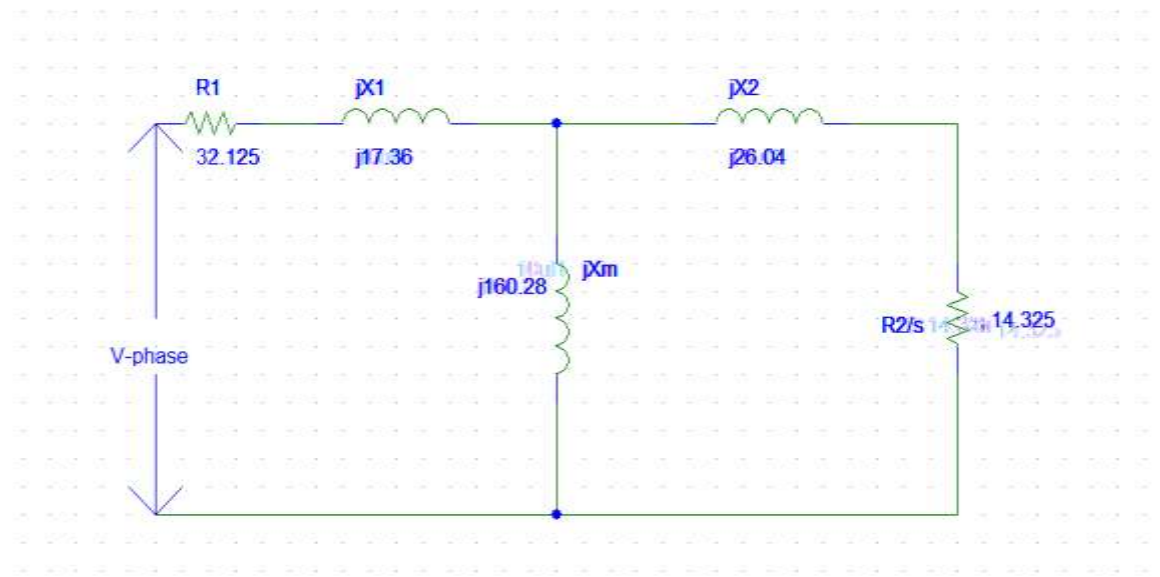


Fig 20: Equiv. Circuit

11. Torque-Speed Characteristics

MATLAB Code for Torque-Speed Characteristics:

```
clc;
close all;
clear all;

R1=32.125;
R2=14.235;
X1=17.36;
X2=26.04;
Xm=160.28;
Vin=415/sqrt(3);
n_sync=1500;
w_sync= 50*pi;

Vth=Vin*(Xm/sqrt(R1^2+(Xm+X1)^2));
Zth=((j*Xm)*(R1+j*X1))/(R1+j*(X1+Xm));
Rth=real(Zth);
Xth=imag(Zth);

slip=(0:0.001:1);
slip(1)=0.0001;
n_m=n_sync*(1-slip);

for i=1:length(slip)
    ind_Tor(i)=(3*(Vth^2)*(R2/slip(i)))/(w_sync*((Rth +
R2/slip(i))^2+(Xth+X2)^2));
end

plot(n_m,ind_Tor,"Color","#4DBEEE", 'LineWidth',2.0); hold
on
title('\bfInduction Motor Torque-Speed Characteristics',
'FontSize', 16);
```

```

grid on;
xlabel('\bfMechanical speed (rpm)', 'FontSize', 13);
ylabel('\bfInduced torque (N.m)', 'FontSize', 13);

plot(n_m(end),ind_Tor(end),'.','MarkerSize', 18, 'Color',
'r');
text(10, 3.2, sprintf('%s = %s N.m', 'Starting Torque',
num2str(ind_Tor(end), '%.2f')), 'FontSize', 12);

pullout_Tor = max(ind_Tor);
pullout_nm = n_m(ind_Tor == pullout_Tor);
plot(pullout_nm,pullout_Tor,'.','MarkerSize', 18, 'Color',
'r');
text(950, 5.4, sprintf('%s = %s N.m', 'Pullout Torque',
num2str(pullout_Tor, '%.2f')), 'FontSize', 12);
text(950, 5.2, sprintf('%s = %s rpm', 'Speed at Pullout',
num2str(pullout_nm, '%.2f')), 'FontSize', 12);

rated_nm = 1400;
indx = find(slip == round(1-(rated_nm/n_sync), 3));
rated_Tor = ind_Tor(indx);
plot(rated_nm,rated_Tor,'.','MarkerSize', 18, 'Color', 'r');
text(rated_nm - 300, 3.1, sprintf('%s = %s N.m', 'Full-load
Torque', num2str(rated_Tor, '%.2f')), 'FontSize', 12);
text(rated_nm - 300, 2.9, sprintf('%s = %s rpm', 'Rated
Speed', num2str(rated_nm)), 'FontSize', 12);

```

Obtained Torque-Speed Characteristics Curve:

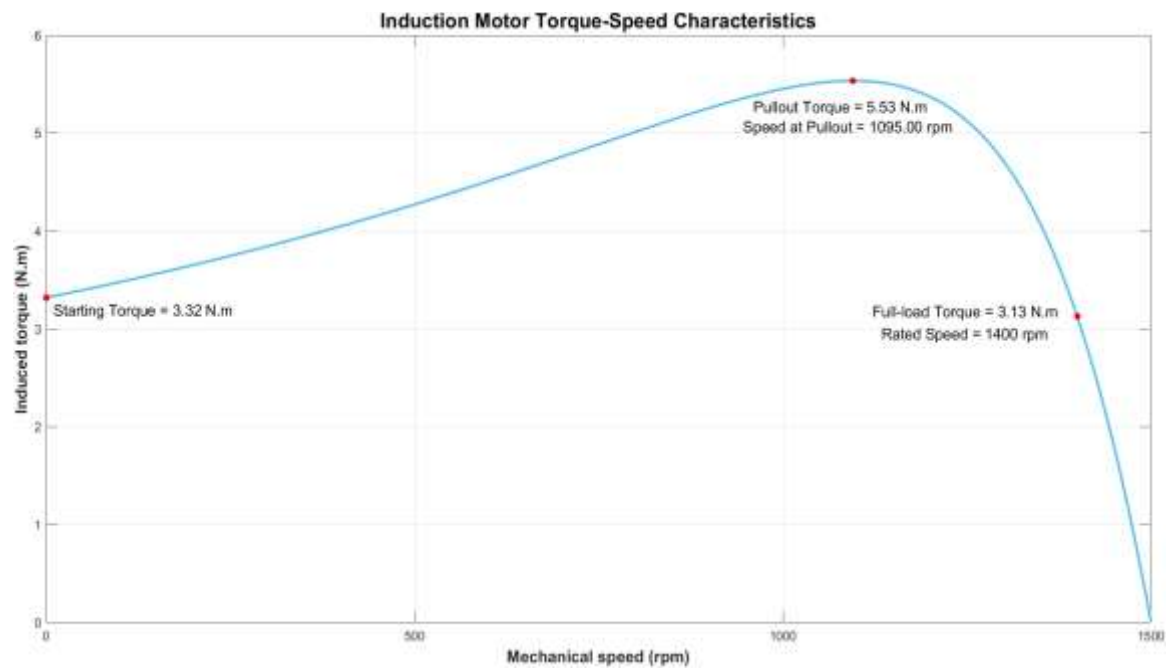


Fig 21: Torque-Speed Characteristics

Summary:

Characteristics	Value
Starting-torque	3.32 Nm
Pullout-torque	5.53 Nm
Speed at pullout-torque	1095 rpm
Torque at rated speed	3.13 Nm

12. Complications and Limitations

Complications:

- Unavailability of professional tools required unavoidable assistance of professional workers.
- Placing coil inside the slots.
- Cutting the old winding.
- Placing insulating cover on slot after placing the coil.
- Mismatch between the calculated wire size and available wire size.
- Lack of essential instruments caused laborious assembling and disassembling of all the parts of the motor.

Limitations:

- The magnetic flux density was not exactly measured.
- Turns per coil is not the same for every coil since turns are counted by weighing the coil.
- Air gap may not be equal everywhere inside the motor which may introduce discrepancy from ideal behavior.
- The insulation class chosen could be more durable but due to high cost a moderate quality class of insulation was chosen.

13. Costing

1	Motor	2400/=
2	Copper Wire, Insulating paper	1000/=
3	Disassemble and reassemble Equipment	400/=
4	Transportation	200/=
Total Cost		4000/=

14. Future Scope of Work

- ✓ The designed motor can be used in real life by mounting its shaft with other machines like pumps, blowers, machine tools etc.
- ✓ Using proper tools and measuring instruments the torque-speed characteristics may be determined experimentally to make the analysis more accurate.

15. Effect on environment

A three-phase induction motor has many mechanical parts such as the stator, the rotor, windings etc. All of these components are composed of certain elements which may be harmful for the environment. Since the industrial revolution has initiated a trend of automation which has no sign of slowing down. Electromechanical equipment will be manufactured at a higher pace in the coming days. Thus, the environmental effect of these equipment should be considered with proper attention. Our project work also requires this investigation. To start off we must identify the primary harmful components used in the project.

Harmful Component	Effect
Copper Wire	Toxic metal
Insulation paper	Non bio degradable
Varnishes	Emit high level of VOC's

Now, to reduce the negative effects of these components we tried to take proper actions for each of the components.

Copper wire:

According to the Agency for Toxic Substances and Disease Registry, industries released an estimated 1.4 billion pounds of copper into the environment in 2000. Copper toxicity can result from chronic or long-term exposure to high levels of copper through contaminated food and water sources. Symptoms of this condition include diarrhea, headaches, and in severe cases, kidney failure. Copper contamination can be made in water, soil, and air. But the soil and food contamination are more relatable to our project. Copper wire is the most cost efficient and electrically superior material for usage. Thus, using copper was unavoidable. But copper mainly affects the environment if it is disposed of in the open. Thus, we recycled the old winding we found initially and planned on recycling the new winding after wearing out of the motor. In this way available copper will not be able to affect the environment.

Insulation Paper:

Non-biodegradable materials are hazardous to the environment because they take a lot of time to get decomposed to its elements. Thus, we used polymer insulation and fish paper insulation. Each insulation is eco-friendly. Thus, there disposal will not negatively affect the environment.

Varnishes:

Varnishes contain VOC's (Volatile Organic compound). When VOC's react with oxygen, they can form 'bad' ozone in the presence of sunlight. This is a contributory factor to the greenhouse effect and a cause of global warming. One study by C.E.P.E. attributes paints and varnishes to causing as much as five percent of all VOC emissions. Varnishes should be used in minimum amount, and it should be used as the instruction manual. Proper protective measure is also essential applying varnishes. Furthermore, after usage the cleaning of brushing instruments is also necessary using proper solvents.

16. Achieved Program Outcomes

1. PO(a) - Engineering Knowledge

- Designing the squirrel cage induction motor required knowledge of motor construction and equivalent circuits.
- Equivalent circuit determination relied on SCC and OCC testing methods.

2. PO(c) - Design/Development of Solutions

- The design of the squirrel cage induction motor involved detailed calculations.
- The motor was constructed according to the calculated parameters.

3. PO(d) - Compare Theoretical and Empirical Results

- The performance analysis of the induction motor involved comparing theoretical calculations with experimental results to validate the design and operation.

17. Conclusion

Our primary goal of the project was to design a three-phase induction motor. The total project work gave us an insight into the construction of the motor. Although, prior to conducting the project we gathered as much knowledge as possible about the construction and the working principle of the motor. But hands-on activities gave a deeper look into the concepts we have learned. Furthermore, finding the equivalent circuit was also a part of the objective. The equivalent circuit helps to make necessary circuit simulations for the motor such as the torque-speed characteristics curve obtained by MATLAB. The winding diagram for the motor also gave us a good understanding of the wire connection. A perfect stator winding connection is crucial for overall motor performance. Some graphics software's were used while drawing the winding diagram. Moreover, while observing the calculation we came to find that many estimated values were not available in practical sense. Thus, adjustments were required. However, some preferences were given only for mitigating negative environment impacts. Such as the wire insulation, recycling the old wire, proper disposal of the non-biodegradable insulation paper etc. Some decisions were made to make the project more economical. This practice helps us to be a better engineer. As engineers need to complete any task efficiently and cost effectively. As a result, some tradeoffs were unavoidable between performance and cost. Additionally, each step of the project has some certain outcomes. All the outcomes we obtained or tried to obtain are part of the 11 program outcomes or POs. Each of the POs have their unique characteristics. We couldn't achieve all the POs in this project. But we tried to mention how, and by which task we achieved the PO's. Additionally, we tried to gather relevant knowledge from various sources, and we tried to give proper reference to every single source we benefitted from. The project is mainly focused on an electrical component but redesigning it and implementing it requires a lot of other subskills. This gives us an idea that only technical knowledge is not enough. Rather other subskills to back the core knowledge is also important.

List of References

1. “DESIGN OF INDUCTION MOTOR.” Accessed: Sep. 08, 2023.[Online]. Available:https://www.jcboseust.ac.in/electrical/images/induction_motor_design.pdf
2. *The Association of Electrical & Mechanical Trades :: Classification of Insulation Systems.* (n.d.-b).<https://www.theaemt.com/resourcesupport/resource-support-resources/general-engineering/classification-ofinsulationsystems#:~:text=Class%20F%20insulation%20consists%20of,higher%20than%20Class%20B%20materials>
3. “Engineering Tables/Standard Wire Gauge - Wikibooks, open books for an open world,” en.wikibooks.org.
https://en.wikibooks.org/wiki/Engineering_Tables/Standard_Wire_Gauge
4. S. J. Chapman, *Electric machinery fundamentals*, 5th ed. New York: McGraw-Hill, 2012.
5. *A textbook of electrical technology, vol. II : AC & DC machines in S.I. system of units.* New Delhi: S. Chand And Company Limited, 2019.
6. P. Mart, “Copper Wire: Resistance, Specific Heat Capacity, and Current Density,” *ThePipingMart Blog*, Dec. 21, 2022.
<https://blog.thepipingmart.com/metals/copper-wire-resistance-specific-heatcapacity-and-currentdensity/#:~:text=Current%20Density%20of%20Copper&text=The%20maximum%20recommended%20current%20density> (accessed Sep. 08, 2023).
7. McFadyen, S. (n.d.). Motor insulation.
<https://myelectrical.com/notes/entryid/154/motor-insulation>